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Staggered Health Policy Adoption: Spillover Effects and Their Implications

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Abstract. This paper investigates the direct and spillover effects on mobility caused by the staggered adoption of stay-at-home orders (SHOs) implemented by U.S. counties to contain COVID-19 at the beginning of the pandemic. We find that mobility in neighboring counties declines by a third to a half as much as in the counties that implement the SHOs. Furthermore, these spillovers are concentrated in counties that share media markets with treated counties. Using directional mobility data, we also find that declines in internal mobility in the neighbor counties account for a much larger proportion of the overall decline in mobility than decreases in traffic originating in the treated counties. Together, these results provide strong evidence that SHOs operate through information sharing and voluntary social distancing. Based on our estimates and a simple model of staggered SHO adoption, we construct counterfactual scenarios that separate the impact of policy coordination from that of adoption timing. We find that staggered implementation of SHOs could yield mobility reductions that are larger than coordinated but delayed SHO adoption.

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Keywords: COVID-19 • smartphone-based mobility data • media markets • nonpharmaceutical interventions • place-based policies • spillovers • stay-at-home orders • voluntary social distancing

1. Introduction

People gather and interact in complex ways. Administrative jurisdictions have the authority to impose local policies that affect their residents. However, proximity and connectedness among contiguous administrative units may propagate beyond their geographic borders the effects of policies that target their populations. Changes in local regulations related to taxation, crime, and public health are especially likely to generate spillover effects beyond the local jurisdiction that adopts them.

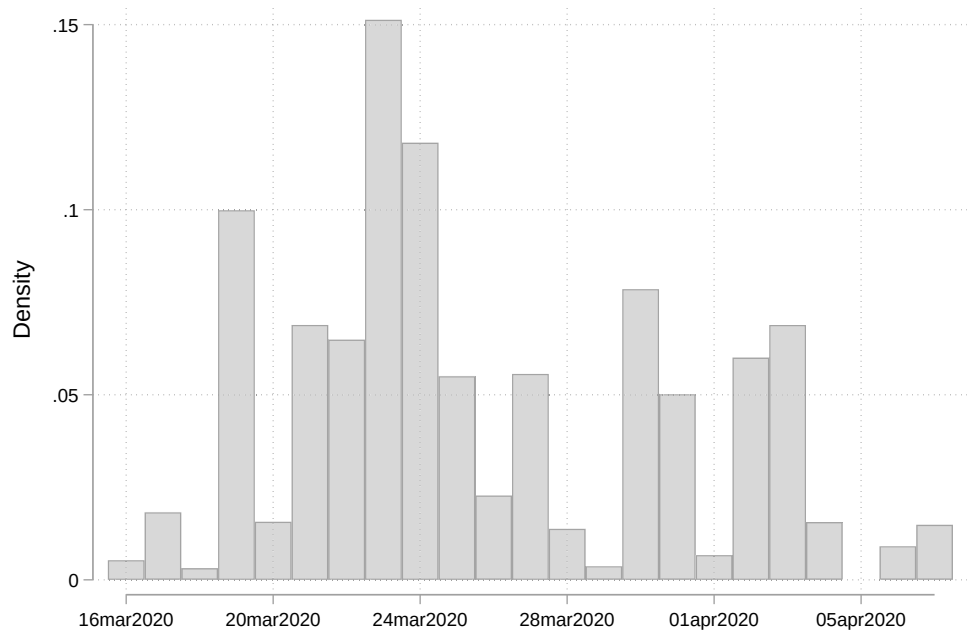
Understanding how such spillovers work is important for several reasons. First, it may help shed light on the mechanisms through which these policies operate. Second, it can broaden our understanding of the total impact of these policies, helping the policymakers to formulate them optimally. Third, properly assessing spillover effects is necessary for an accurate quantification of direct effects in adopting jurisdictions. In this paper, we investigate the direct and spillover effects of one particular public health policy—the stay-at-home orders

(SHOs)—adopted to slow the spread of COVID-19 in the United States in the spring of 2020.¹

In the United States, nonpharmaceutical interventions (NPIs), such as SHOs, were implemented by local jurisdictions in a staggered manner over time and across geographies without national coordination. Figure 1 shows the distribution of SHO adoption dates across U.S. counties at the onset of the COVID-19 epidemic, weighted by county population. The earliest adopters were counties in the San Francisco Bay area, shutting down on March 15, 2020. Over the next four weeks, much of the country adopted SHOs, with parts of South Carolina and Missouri that were still open shutting down in the first week of April. The histogram reveals a substantial dispersion in adoption dates. With the staggered implementation of SHOs across space and over time, there is ample scope for spillovers.

Theoretically, spillovers in policy implementation can occur as agents' actions and behaviors in a jurisdiction affect directly or indirectly agents outside the implementing location, for example, through peer effects,

Figure 1. County Implementation of Stay-at-Home Orders



Notes. The histogram shows the distribution of adoption dates in all U.S. counties that adopted an SHO between March 16, 2020, and April 7, 2020. Observations are weighted by county population. Four hundred sixty-one counties representing 3.9% of the population never adopted an SHO in the March–April 2020 period and are not included in this histogram.

strategic interactions, and externalities. This is particularly important when jurisdictions have the institutional latitude to act independently but are connected, which is the case in the decentralized government system of the United States, where the median county has 15% of its traffic originating outside its borders. As a result, policy coordination among contiguous or connected jurisdictions is generally considered beneficial.

America’s decentralized approach to the COVID-19 fight in the early phase of the pandemic has been particularly criticized, in public discourse in the media, in policy debate, and in the extant literature. A poll by *The Economist* magazine in April 2020 showed that 61% of people believed that President Donald Trump should issue a national stay-at-home order. Anthony Fauci, National Institute of Allergy and Infectious Diseases (NIAID) Director at the time, in an interview with *FiveThirtyEight*’s “PODCAST-19” on July 9, 2020, stated, “So I think you’d have to make the assumption that if there wasn’t such divisiveness, that we would have a more coordinated approach” (Fauci 2020). The Organisation for Economic Co-operation and Development (OECD), in a report on the territorial impact of COVID-19 and lessons for crisis management across levels of government (OECD 2020, p. 2), also concludes: “In the United States, (...) regionally differentiated (COVID) impact calls for a territorial approach to policy responses on the health, economic, social, fiscal fronts, and for very strong intergovernmental coordination”—

parentheses added. In the extant literature, papers advocating for a coordinated adoption of SHOs based on various rationales include, among several others, Murray (2021), Holtz et al. (2020), Zhao et al. (2021), Chandrasekhar et al. (2021), and Shi et al. (2022).²

However, in theory, the direction of the spillover effects from *staggered* SHO adoption is ambiguous. On the one hand, the intensity of social interactions in a neighboring county could increase if nonbinding restrictions were to induce residents of adopting counties to shift their leisure activities to neighboring counties, thus generating “negative” spillovers. On the other hand, contacts and interactions could also decrease through voluntary social distancing in response to a nearby treated county’s adoption of more stringent SHOs because of information sharing and transmission and peer effects that raise awareness about the increasing risks of an infectious disease epidemic—the spillovers could be “positive.” Hence, it is an important empirical question whether the staggered SHO implementation across U.S. counties entailed spillovers, in which direction spillovers predominantly went, and what implications they may have for the design of desirable location-based health policy interventions intended to stop infectious disease spread.

In this paper, we estimate the direct and spillover effects from staggered SHO implementation in the United States in March and April 2020 and investigate the mechanisms driving them. We then use our

estimates to counterfactually evaluate alternative SHO implementation regimes, distinguishing coordinated from early adoption. We propose an identification strategy based on comparing adjacent counties in triplets of contiguous counties in which one county adopts an SHO at least four days before the others. The approach compares the treated county to its neighbor and to the neighbor's neighbors, which we term *hinterland* counties. It critically differs from strategies previously proposed in the literature in that it addresses the risk of contamination of the estimated effects from simultaneous (or near-simultaneous) adoption in the neighboring counties.³

A first set of estimation results illustrates that spillovers from staggered SHO adoption are sizable and go in the same direction as the direct effects. In response to an SHO, traffic in neighboring counties drops by about a third to a half as much as traffic in adopting counties, when compared with hinterland counties. Next, we show that the estimated spillover effects are more pronounced when adopting and neighbor counties share local news information in Nielsen's common media market areas (designated market areas (DMAs)). This second set of results suggests the presence of an information channel through which local SHOs send a signal to residents of neighboring counties about the severity of the epidemic that induces mobility reduction, consistent with behavioral models of COVID diffusion. Finally, we also show that spillovers happen both because residents of the early-adopting counties reduce rather than increase visits to nonadopting neighbors (i.e., they cause an external traffic reduction) and especially because the neighbor's own residents change their behavior in response to the adopting county's SHO by voluntarily reducing mobility within their own county (i.e., they cause a reduction in internal traffic). Moreover, we show that the contribution of this internal traffic channel is about twice as large as that of the external traffic. This third set of results corroborates the importance of voluntary social distancing.

Our results have an important health policy implication. The critical finding that the estimated spillovers are sizable and in the same direction as the direct effects opens up the possibility that the staggered implementation of health policies may be preferable to simultaneous *but* delayed adoption. We explore this possibility in a simple calibrated model of staggered and simultaneous SHO adoption and in counterfactual analysis based on our reduced-form estimates. The results illustrate that this can indeed be the case, qualifying the widespread view that a nationally coordinated policy response would have accomplished a greater reduction in mobility and social distancing regardless of when implemented. Indeed, we find that coordination *and* early adoption always achieve greater mobility reduction. But delayed coordination alone could be dominated by

staggered implementation as our counterfactual analysis shows. The intuition is that, with positive spillovers, staggered adoption entails additional rounds of treatments that contribute to reducing overall mobility both by lowering external traffic to neighboring counties and through information sharing about the epidemic risk.

The rest of the paper is organized as follows. In the next section, we discuss the paper's contribution in the context of the literature. Section 3 presents the data. Section 4 discusses our identification strategy and assumptions. Section 5 reports the estimated direct and spillover effects and then investigates the mechanisms that could drive the spillover effects that we document. Section 6 presents our counterfactual policy evaluation analysis that highlights the differential impact of coordination and timing of policy adoption in the presence of positive spillovers. The last section concludes. Details of the empirical and counterfactual analyses are reported in an Online Appendix.

2. Related Literature

This study is at the intersection of three main strands of the literature: research on the econometric implications of spillovers, studies on place-based policies, and the literature on COVID-19 policy interventions and their effects on mobility, economic activity, and health outcomes, including the desirability of coordinated policy adoption of SHOs.

Several papers have shown that spillovers can lead to biased estimates of the direct effects of policy interventions if they are not properly accounted for. For example, Cao and Dowd (2019) show that spillovers can bias the estimation of direct treatment effects in the tobacco control program study in Abadie et al. (2010). Kalenkoski and Lacombe (2013) demonstrate that analyses of minimum wage changes, such as the seminal work of Card and Krueger (1994), can suffer from bias when spillovers are ignored, especially when contiguous counties are selected as controls because of spatial heterogeneous trends, as in Dube et al. (2010) and Neumark and Kolko (2010). Hanson and Rohlin (2013) show how using nearby areas as controls to evaluate policies, as analyzed for labor and banking deregulation policies in Holmes (1998) and in Huang (2008), and crime prevention in Blattman et al. (2021), can induce upward biases if the spillovers are negative. Many other studies report results that remove neighboring locations to alleviate the concern of spillover such as, for example, Kline and Moretti (2014). This study adds to the literature by proposing a jurisdiction-level identification strategy that can be applied in other domains of policy analysis.

Our work contributes to the literature on place-based policy evaluation. Several studies have documented spillovers from place-based policies stemming mainly from mobility, similar to our work. These include

spillover effects when local jurisdictions compete in welfare policies with nearby locations to contain population losses (Baicker 2005, Isen 2014), when implementing jurisdictions shift negative behaviors targeted by the adopted policy to neighboring locations (Blattman et al. 2021), and when neighboring jurisdictions see policy implementation in nearby locations as a signal that similar policies might be implemented locally in the future (Galletta 2017). Our work contributes to this literature by proposing a conceptual framework for the estimation and counterfactual analysis of the spillover effects that is novel in this domain.

The paper contributes to the vast literature on COVID-19 policy interventions and their effects on mobility, economic activity, and health outcomes, and particularly the evaluation of nonpharmaceutical policies.⁴ Goolsbee and Syverson (2021) evaluate how SHOs affect mobility within commuting zones across state and county boundaries, as in our analysis of internal mobility. Similarly, Lin and Meissner (2020) consider SHOs at the state level and how their impact changes with and without including nearby states' SHOs. Murray (2021) finds that SHOs can lead to significant mobility declines, but their effects are temporary, thus calling for coordinated action. None of these studies control for spillover effects on neighboring jurisdictions. We estimate both direct and spillover effects of county SHO adoption, as well as their short-term dynamics, and we also decompose the spillovers into external and internal mobility, highlighting a link from internal mobility to information sharing and transmission in common media markets that emphasizes the importance of voluntary social distancing.

Loewenthal et al. (2020) also find that the time at which social distancing was initiated is highly correlated with the number of deaths whereas the lockdown strictness or its duration is not, and conclude that faster adoption of the lockdowns could have delivered greater mobility and contact reduction. The results of our counterfactual analysis are consistent with these conclusions. In related work, Berry et al. (2021) estimate that shelter-in-place in the United States had no significant causal impact on social distancing behavior or the economy because people had already changed their behaviors before the introduction of shelter-in-place orders. Although we find that mobility started to decline before SHO adoption, we show that postadoption trends significantly diverge and that SHOs worked through information sharing and transmission and voluntary social distancing.

Several studies raise the possibility of NPI spillovers from a theoretical perspective. For example, Bethune and Korinek (2020) theoretically show that externalities play an important role in limiting voluntary self-isolation in response to COVID-19 as the impact of individual choices on the aggregate infection risk is not

internalized. A similar mechanism can apply to local authorities deciding on NPI implementation with respect to neighboring jurisdictions, with a potential negative spillover effect from residents of the late adopter to the early adopter. Cui et al. (2020) model how the policy of one state can influence the incentives that other states face to adopt. If enough states engage in social distancing, they will tip others to do the same by shifting the Nash equilibrium with respect to the number of states engaging in costly social distancing. Similarly, Beck and Wagner (2020) develop a theoretical model to show that uncoordinated interventions can have spillovers across countries. Although the main contribution of the paper is empirical, we also set up a simple model of NPI adoption in which spillovers in the same direction as the direct effects can contribute to rather than detract from the goal of reducing mobility and interactions.

Chandrasekhar et al. (2021) examine three adoption regimes in a network model of infectious disease contagion to understand the impact of coordination or lack thereof in the implementation of NPI policy. They consider adoption by a central authority or by separate jurisdictions that can either react only to internal infection rates or react to both internal and external infection rates. Although this study derives conditions under which staggered implementation of NPIs can deliver a socially efficient outcome like in our model, the paper concludes that in practice coordination in both the design and execution of policies across jurisdictional boundaries is desirable.

We set up a model in which the timing of implementation matters and staggered implementation may or may not be desirable. However, once we calibrate it to the data by using our estimates, we find that staggered policies could be more desirable than simultaneous adoption under a wide set of parameters. Our policy conclusions are similar to those in Crucini and O'Flaherty (2021). However, the mechanism in that study is different, relying on the heterogeneous distribution of the economic cost and the health benefits of SHO adoption as in Eichenbaum et al. (2021) or Shi et al. (2022) and several other behavioral epidemiology models with economic outcomes.

Recent work has examined spillovers resulting from staggered *lifting* of mobility restrictions. Chang et al. (2022) find that geographic mobility from counties subject to more restrictions to counties in which restrictions are lifted increases. As this work focuses on removing, rather than imposing, stricter mobility measures, our results are not directly comparable. However, the importance of signaling (information) that is contained in policy changes to mobility restrictions for individual behavior is underscored in both studies.

Our work is most closely related to work by Holtz et al. (2020) that empirically examines the possibility of

mobility spillovers in SHO implementation like in this paper. Holtz et al. (2020) estimate the direct and spillover effects of county SHO adoption on mobility, finding positive spillovers and concluding that coordinated (i.e., simultaneous or contemporaneous) adoption would have achieved greater mobility decline. In this paper, we focus on staggered policy adoption as observed in practice during March–April 2020 and contrast this to counterfactual coordinated adoption.⁵ We examine the case where one county adopts a stay-at-home order but the neighboring county does not, at least for a period of four days before and three days after the implementation. Focusing on spillovers from staggered SHO implementation permits separating the mobility impact of coordinated policy adoption from that of the timing of implementation. This is a critical distinction that leads to a new and complementary conclusion, suggesting that a *staggered* implementation might achieve bigger mobility declines than *delayed* coordinated adoption. Like Holtz et al. (2020) (and Zhao et al. 2021), this paper also concludes that coordinated *early* adoption is the preferable regime.

Still, our study and the work of Holtz et al. (2020) also differ along a couple of important dimensions. In econometric terms, focusing on staggered versus (early or delayed) coordinated policy adoption requires imposing an additional identification restriction compared with Holtz et al. (2020), namely an implementation timing gap between the treated and neighbor county (or connected county in their setting). Additionally, the inefficiency associated with lack of coordination in the model in Holtz et al. (2020) is not due to positive spillovers, but to a free-riding problem in the intensity of adoption of costly policy implementation that is not necessarily in the data.

Staggered adoption is the policy regime actually observed in the data. In our model, staggered adoption can be preferable to coordinated but delayed adoption because of the additional round of treatments that staggering entails. Therefore, the analysis of staggered implementation not only reduces further the risk of contamination of the spillover estimates from simultaneous or near-simultaneous adoption in neighboring counties, but it is also relevant on its own because it permits us to separate the contribution to mobility reduction of simultaneous adoption from that of implementation timing.

3. Data

Our empirical analysis of mobility in response to SHOs relies on smartphone location data collected by SafeGraph.⁶

SafeGraph location data come from signals, or “pings,” that identify the location of a particular smartphone at a moment in time. In January 2020, SafeGraph data contained information on more than 45 million devices, with an approximate coverage of 16% of all

smartphones in the United States. Tracking device pings, the data identify visits to more than 5 million points of interest (POIs). The data cover all areas of the United States, with similar coverage in and out of the county sample, with 0.113 devices per capita tracked in counties in sample and 0.107 out of sample, during March and April 2019.⁷ Couture et al. (2021) show that smartphone location data provide an accurate representation of mobility patterns in the United States and can match patterns reported in conventional survey data. For all indicators, the resulting sample comprises daily observations at the county level from March 1 to April 30, 2020, and the same period in 2019.

We aggregate visits to POIs at the county level and daily frequency by using the *SafeGraph Weekly Patterns Data set*, which at the time of access was released weekly. This data set reports total visits to individual POIs together with the POI’s location. It includes the total weekly visits to each establishment, as well as a vector with the distribution of these visits across days of the week. We construct our mobility measures by aggregating visits to POIs classified as restaurants, retail, and an aggregate of all nonmedical establishments. Our POI classification comes from the *SafeGraph Core Places Data set*.⁸ Medical establishments include the following categories (called *Top Categories by SafeGraph*) that we exclude from the construction of our total mobility indicator: General Medical and Surgery Hospitals, Individual and Family Services, Medical and Diagnostic Laboratories, Physicians’ Offices, and Outpatient Care Centers.

Specifically, the mobility variables that we construct from the Weekly Patterns Data set are (i) total visits to all nonmedical establishments, a variable that captures interactions in consumption and leisure activities; (ii) visits to restaurants; and (iii) visits to retail locations.

We also construct social distancing measures from the *SafeGraph Social Distancing Data set*. This data set reports daily visits aggregated at the Census Block Group (CBG) level. For each device, SafeGraph tracks the total time spent at home during the day and reports the median across all devices in a given CBG. SafeGraph also defines part-time and full-time work behavior as devices that spend more than six or three hours at one location, respectively, other than the device’s home base (the geohash-7) during the 8 a.m.–6 p.m. local time period.

From this database, we use (iv) the number of hours spent at home, whose change can be interpreted as an indicator of social distancing, and (v) the fraction of devices that stay completely at home, without leaving the geohash-7 location in which the home is located, which is an indicator of overall social distancing. We then calculate (vi) the fraction of devices working away from home, which is a proxy for interactions at work, by adding the median number of devices classified as working part-time and full-time and dividing by the

total number of devices in a CBG. To aggregate CBG data at the county level, we take a device count-weighted average across census blocks within a county.

Besides the total visits to a CBG, SafeGraph's Social Distancing Data set also includes a vector with the distribution of these visit origins.⁹ Using this information, we build external (i.e., between-county) daily mobility flows as well as a measure of internal traffic flow (visits to a county by devices residing in that same county) that we use in our analysis of mechanisms. These measures may be affected by SafeGraph's privacy rules when we consider pairs of counties with low traffic between them. Online Appendix Table A.2 shows the average number of visits across and within county borders in this sample. We discuss how we consider this censoring in Section 4.2.

Furthermore, we also use directional traffic flows to measure connectedness between any two origin (o) and destination (d) counties as follows:

$$C_{od} = \frac{\omega_{od} + \omega_{do}}{\sum_d (\omega_{od} + \omega_{do})}, \quad (1)$$

where ω_{od} is the total number of visits originated from location o traveling to destination d , and ω_{do} the total number of visits originated from location d traveling to destination o . To measure connectedness before the pandemic, we add up weekly traffic between any two locations in March and April in 2019. We use this measure of connectedness as a selection criterion for control (hinterland) counties.

We focus on SHOs throughout the paper. SHOs are advisory notices that people should shelter at home except for essential reasons. These notices have also been referred to as shelter-in-place orders or lockdowns. We focus on SHOs because they were the most widely adopted NPIs with substantial variation in adoption both over time and across counties. The data set on the county-level implementation of SHOs merges county-specific and state-specific adoption dates from Keystone Strategy, a consulting firm, and the Johns Hopkins Coronavirus Resource Center data set assembled in Killeen et al. (2020).¹⁰ Figure A.1 in the Online Appendix plots the adoption dates that we utilize.

We use weather conditions such as temperature, precipitation, and snowfall as controls in the analysis. We obtain the weather variables from National Oceanic and Atmospheric Administration (NOAA) weather stations matched to the closest county based on proximity to its centroid and data availability. When multiple weather stations are available, we use the one with the best coverage. Finally, we use population data from the 2018 American Community Survey (ACS) to weigh observations in some robustness specifications.

4. Methodology

Our main goal is to evaluate the effects of the staggered introduction of NPIs on mobility in U.S. counties.

Mobility is a good proxy for contacts and interactions, which are the fundamental drivers of infectious disease transmission (Adda 2016, Murray 2021) and whose reduction has been the main objective of place-based health policy interventions. In an ideal research design, there would be sets of ex ante identical counties. Simultaneously in each set, one county would be randomly treated with an SHO. Subsequent differences in outcomes could then be unambiguously attributed to the SHO. Changes in outcome variables in other counties would measure spillovers, whereas changes in outcome variables in the treated county would measure direct effects.

However, because SHOs are not randomly implemented, a simple comparison of mobility before and after SHO adoption across counties could attribute the estimated impact to a specific county's SHO, whereas, in fact, it could be caused by the previous adoption of another county's policy, a regional trend, a seasonal pattern given geography, differences in economic activity, the perceived risks of the epidemic and the costs of shutdowns, or other unobservable factors. As counties that are geographically close are more likely to experience similar economic or epidemic trends, as well as similar geo-specific shocks that are unobservable to the researcher, we focus on studying sets of counties that are close to each other.

4.1. Identifying Direct and Spillover Effects

To identify direct and spillover effects of staggered implementation of NPIs on mobility, we adopt an identification strategy based on comparing *adjacent* counties in triplets of geographically *contiguous* counties ("contiguous triplets"), in which one county adopted an SHO *earlier* than others. We call this research design identification by contiguous counties for brevity.

The critical identifying assumption is that contiguous counties share unobserved time-varying characteristics. Additionally, we impose the restriction that counties do not implement policies on the same day. As a result, residual variation in ex post outcomes can be attributed to the SHO treatment of the adopting county in the triplet. By restricting the analysis to contiguous triplets in which the implementing county adopts the SHO *before* the others, we avoid confounding spillovers that arise from *staggered* implementation with those that might arise from *simultaneous* adoption within the triplets.¹¹

Our research design relies on comparing outcomes in counties that implemented an SHO with adjacent and nonadjacent nearby counties that did not adopt an SHO for at least four days after the treated county's implementation. Following Huang (2008), we identify sets of three counties, or triplets. The *treated* county in the triplet adopts the SHO. The *neighbor* county shares a border with the treated county and does not adopt an SHO until at least four days after the treated county's adoption

date. The four-day adoption gap permits us to isolate the impact of adoption in the focal county from simultaneous or lagged implementation effects in the neighboring county.

The hinterland or *control* county is drawn from the group of counties that share a border with the neighbor county but do not share a border with the treated county, nor share a border with any other county that adopts an SHO for at least four days after the treated county's adoption date. By focusing on studying the effects of staggered as opposed to simultaneous adoption in this setup, we can cleanly identify the impact of NPI implementation on mobility in the treated and the neighbor counties. We conduct numerous robustness analyses on various subsets of this sample of triplets, including sensitivity to the imposed four-day implementation gap restriction.

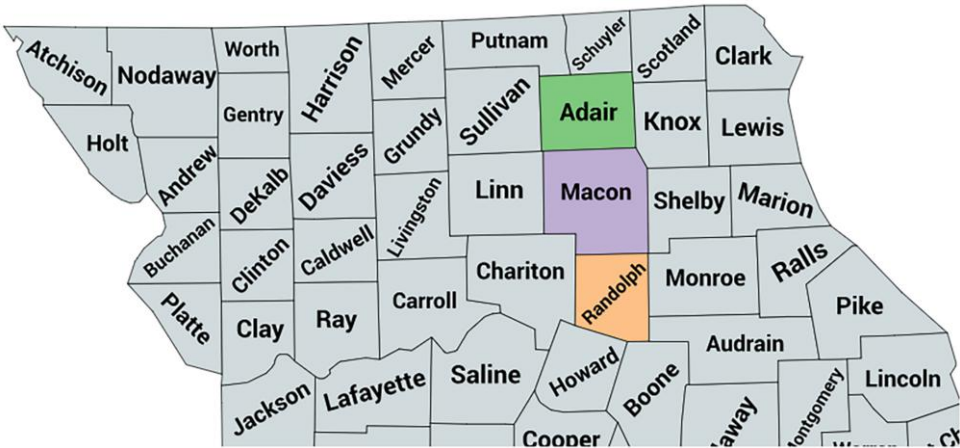
To fix ideas, Figure 2 zooms in on one such triplet in the north of Missouri. Randolph County is a treated county because it adopted a stay-at-home order on March 25, 2020. Macon County did not adopt a stay-at-home order until April 6. Yet, because it shares a border with Randolph County, the mobility and social interaction of its residents could be subject to spillover effects from Randolph County's order on March 25. We call Macon a neighbor county. Adair County borders Macon County, but it does not border Randolph County or any other county that adopted a stay-at-home order on or before March 29. It also had not adopted an SHO by March 25, and did not do so until April 6—long after the four-day threshold. As a result, we select it as the control county and we call it a hinterland county. It is “close enough” to Randolph and Macon Counties to share their economic and epidemic characteristics, but is “far

enough” from Randolph County to be minimally affected by its SHO.¹² To the extent that outcomes change in Randolph County more than in Adair County after March 25, we interpret this as a direct effect of Randolph County's order. To the extent that outcomes change in Macon County more than in Adair County after March 25, we interpret this as a spillover effect of Randolph County's order.

To construct our sample of county triplets, we begin with a list of all adjacent county pairs from the U.S. Census Bureau. We then eliminate adjacencies over large bodies of water (e.g., counties in Wisconsin bordering counties in Michigan across Lake Michigan). This step leaves us with 18,870 county pairs. Next, we retain only those pairs in which one county adopted an SHO at least four days after the other if it did at all. The choice of time interval between adoption days involves a trade-off. Selecting a small number of days results in a narrower window over which we can identify a staggered spillover effect because the neighbor county may receive direct treatment right afterward. This not only increases the risk of contamination from neighbor adoption but may also misstate the total effect if it takes several days to unfold. Choosing a wider time window (a higher number of days) results in fewer triplets, significantly reducing power. Taking both concerns into account, we chose four days. In the Online Appendix, however, we repeat the analysis for three-day and five-day intervals and confirm that our main results are robust for all outcome variables. This second step leaves us with 1,063 county pairs. The early adopter is the *treated* county. The late adopter, if it adopted at all, is the *neighbor* county.

We then identify hinterland counties among the counties adjacent to the neighbor that have not implemented

Figure 2. (Color online) Contiguous County Triplet Example



Notes. This map shows a subset of counties in Missouri with one of the county triplets we select for our empirical analysis as an example. Randolph County is a treated county because it adopted an SHO order on March 25, 2020. Macon County did not adopt an SHO until April 6, 12 days later. We call it a neighbor to the treated county, and it is potentially subject to spillover effects from Randolph County's order. Adair County borders Macon County, but it does not border Randolph County or any other county that adopted an SHO on or before March 29, nor did it adopt an SHO itself by that deadline. Thus, Adair is selected as the control county, and we call it hinterland.

an SHO and that do not qualify as neighbors themselves. For some pairs, no such county exists—all of the neighbor's adjacent counties either adopted an order within four days or border others that did so. As we want to identify direct and spillover effects not contaminated from simultaneous SHO implementation in contiguous counties, we eliminate these pairs. For other pairs, there are one or more counties that meet this criterion to be a hinterland. In these cases, we select the one that is “not connected” to the treated county, in terms of having the least traffic between the two counties prepandemic, in March–April 2019. This third step leaves us with 856 triplets of contiguous counties.

Additionally, to ensure that neither the neighbor nor the hinterland county has been affected by other counties' prior orders, we also remove triplets where those counties border an earlier adopter. Consider the following example based on Figure 2. Suppose that Ralls County on the eastern border of the state adopts an SHO on March 29, whereas Monroe, its western neighbor, and Shelby, Monroe's northern neighbor, wait until April 6 to adopt their orders. This would make Ralls–Monroe–Shelby a candidate triplet of treated, neighbor, and hinterland counties, respectively. However, Monroe also borders Randolph County, which—as the map shows—adopted on March 25, prior to Ralls' hypothetical March 29 adoption. To the extent that spillovers occur, Monroe may have already experienced a round of treatment. Therefore, we do not include the Ralls–Monroe–Shelby triplet in our sample. The same exclusion would occur if Shelby, the hinterland county in the candidate triplet, bordered Randolph. However, if the Ralls adoption took place on the same day as Randolph's—March 25—then both triplets would remain in the sample. Excluding triplets based on this additional criterion leaves us with 550 triplets, which form the basis of our analysis.

Because our initial data set includes all pairs of counties in the United States, the same treated county could have several adjacent counties, none of whom have themselves adopted an order and all of whom border their own suitable hinterland county. In these cases, we construct several triplets that all share the treated county but have different neighbor and hinterland counties. Our final sample of triplets includes 256 unique treated counties. Similarly, the same nonadopting county could share a border with multiple other counties all of which adopt on the same day. In these cases, we will have several triplets that have the same neighbor county but different treated counties. Online Appendix A.17 contains the list of all counties used in the analysis, and their designation as treated, neighbor, or hinterland. In Online Appendix Section A.2, we consider alternative sample constructions and show that our results are robust.

The final sample covers 890 counties, about a quarter of all continental U.S. counties, containing just over a

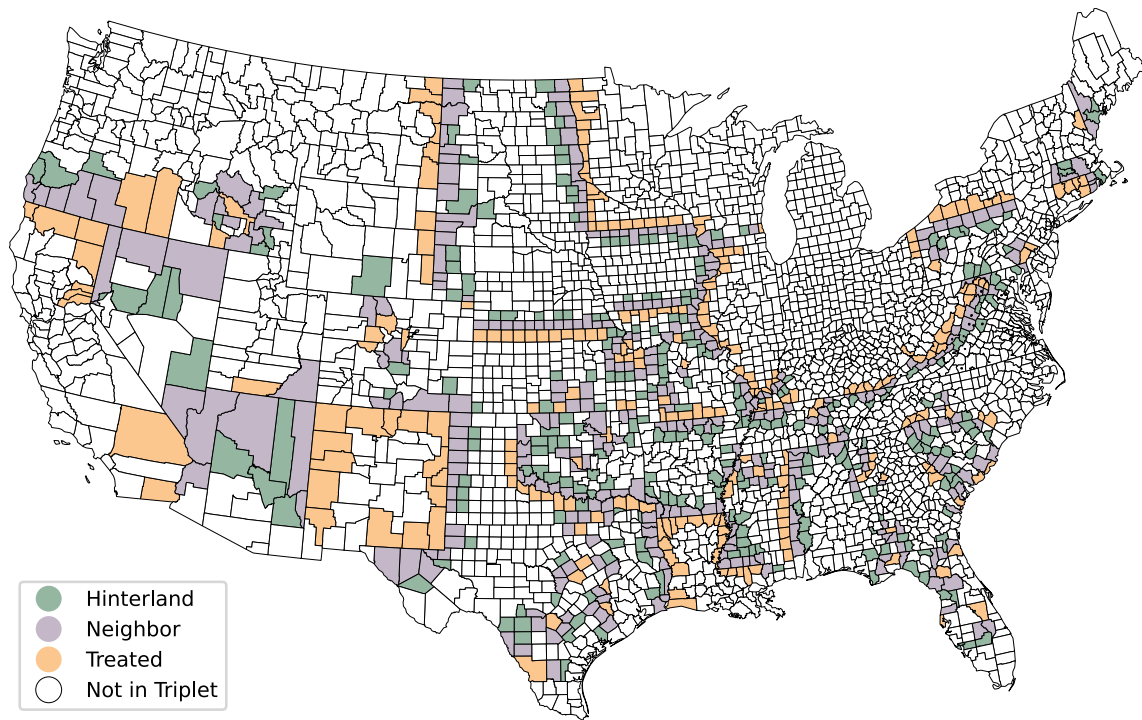
fifth of the total U.S. population, spread throughout the country. Figure 3 shows each treated, neighbor, and hinterland county in the final sample. The triplets making up our final sample often lie along state borders, but not always. Table A.1 in the Online Appendix summarizes characteristics of the selected 890 counties based on the 2018 American Community Survey, along with the comparable characteristics for those counties that are left out of our sample in the triplet selection procedure. The counties in our sample are more rural: density and transit scores show economically significant differences.¹³ Our sample of selected counties is otherwise broadly comparable to the sample of excluded ones. Other statistically significant differences at conventional levels reflect tightly estimated but small differences. For example, the counties in our sample have a lower median income and a higher share of households below the poverty line. They have a higher share of African Americans and a lower share of Hispanics. These differences highlight a trade-off in our analysis between the risk to external validity and the risk to identification.

Table 1 reports summary statistics for socio-demographic characteristics and all mobility measures in our sample of triplets. The first two columns show the unconditional mean and standard deviation across all counties in the triplet sample. In next three columns, we report the means (with standard deviations in parentheses) across the three counties separately. The last two columns report mean differences between neighbor and hinterland, and treated and hinterland (with standard deviations in parentheses).¹⁴ The hinterland and neighbor counties are on average less populated than the treated counties, and they have a lower proportion of individuals aged over 65. Treated counties are more densely populated and more urban than the neighbor and the hinterland counties. However, there are no substantial differences in the proportion of minorities and in the average economic conditions as measured by median incomes and poverty rates. Treated counties have slightly more college graduates and have more intensive care unit (ICU) beds. On average, hinterland and neighbor counties are quite similar on most characteristics.

On average, devices in the treated and neighbor counties visited more establishments in 2019 relative to hinterland counties. This is consistent with a younger population and a higher transit score. There are no large significant differences in the share of devices visiting workplaces and hours spent at home across the three subsamples. In the last row, we show the proportion of treated, hinterland, and neighbor counties that adopted SHOs during our observation window. By construction, all treated counties implemented an SHO. About 70% of the neighboring and hinterland counties also did so eventually.

Given this sample of contiguous county triplets, we estimate a difference-in-differences regression comparing treated (denoted T) to neighbor (denoted N)

Figure 3. (Color online) Contiguous County Triplet Map (Four-Day Implementation Gap)



Note. The map reports the final sample of treated (orange, light-shaded), neighbor (purple, medium-shaded), and hinterland (green, dark-shaded) counties.

counties, both relative to the hinterland county (denoted H). The *treated* county captures the direct effect of the SHOs, whereas the behavior of the *neighbor* county captures the spillover effect. The regression specification is as follows:

$$y_{c(pr)t} = \psi_{c(pr)} + \eta Post_{pt} + \beta_T(T_{pr} \times Post_{pt}) + \beta_N(N_{pr} \times Post_{pt}) + X_{c(pr)t} + \epsilon_{prt}, \quad (2)$$

where p denotes the county triplet, $r \in \{T, N, H\}$ denotes the county's role in the triplet, $c(pr)$ is the county in the triplet and its role, and $t \in [-4, 3]$ denotes the number of days since the treated county in the triplet adopted its SHO. The unit of observation is a triplet-role-day indexed by prt . To the extent that some counties appear in multiple triplets, they are included multiple times in the sample and are indexed by c , which is a function of the triplet p and the role r . Here, $y_{c(pr)t}$ denotes any of the outcome variables we consider. T_{pr} and N_{pr} are two dummies equal to one if the county is “treated” ($r = T$) or “neighbor” ($r = N$) in the triplet, respectively. $Post_{pt}$ is a dummy equal to one after the treated county in the triplet p implements the order. $X_{c(pr)t}$ is a county- and time-specific vector of weather-related controls—temperature, precipitation, and snowfall. We include county fixed effects in all specifications, $\psi_{c(pr)}$, obviating the need for fixed effects at the triplet or role level, because they would be collinear with the county fixed

effects. To take into account the heterogeneous size of the county triplet, we weigh observations by total population in the triplet as recorded in the 2018 American Community Survey. This implies that the estimated coefficient estimates the local treatment effect for an average county resident.

We impose an eight-day gap around the SHO implementation date for all triplets—four days before the SHO date, the day the SHO is implemented in the treated county, and three days after SHO implementation. Crucially, every treated-neighbor county pair is deterministically matched to a hinterland (control) county in the same triplet on the same date. We create a balanced panel around each triplet's implementation date and stack the panels together. We then estimate a two-way fixed effects model in a difference-in-differences setup. This setup is similar to Cengiz et al. (2019), who use an eight-year window around the implementation date of state-level minimum wage increases to estimate the effect of minimum wages on the labor market. It is discussed in Wing et al. (2024), who term this a trimmed stacked difference in differences. As Gardner (2022), Wing et al. (2024), and Roth et al. (2023) conclude, this setup avoids the issues with estimating difference-in-differences models using two-way fixed effects in staggered treatment setting, as raised by Goodman-Bacon (2021) and others.¹⁵

To cluster the standard errors, we place county triplets into clusters based on the state and date of

Table 1. County Sample Characteristics

Variable	Unconditional		Subsamples			Differences	
	Mean	St. dev.	Hinterland	Neighbor	Treated	Neighbor	Treated
Total visits	6.861	3.671	6.148 (0.1722)	7.132 (0.1993)	7.231 (0.2595)	0.9845*** (0.2634)	1.083*** (0.3114)
Restaurant visits	1.620	1.295	1.388 (0.04935)	1.735 (0.08429)	1.703 (0.07508)	0.3470*** (0.09768)	0.3151*** (0.08985)
Retail visits	2.342	1.429	2.082 (0.07242)	2.399 (0.07398)	2.534 (0.1024)	0.3178*** (0.1035)	0.4523*** (0.1254)
Hrs home	9.236	1.226	9.137 (0.07681)	9.325 (0.06623)	9.215 (0.07018)	0.1880* (0.1014)	0.07801 (0.1040)
Share completely home	29.98	4.391	29.68 (0.2655)	29.61 (0.2286)	30.83 (0.2729)	−0.06907 (0.3503)	1.153*** (0.3807)
Share work	14.60	2.201	14.50 (0.1346)	14.77 (0.1188)	14.47 (0.1303)	0.2703 (0.1795)	−0.03530 (0.1873)
Population	77,124	198,239	38,849 (5,137)	68,161 (8,535)	130,351 (18,473)	29,311*** (9,961)	91,502*** (19,173)
Share 65+	19.55	4.750	20.10 (0.2953)	19.26 (0.2365)	19.39 (0.3084)	−0.8364** (0.3783)	−0.7042* (0.4269)
Share Black	11.46	16.25	11.73 (1.028)	10.74 (0.8226)	12.18 (1.022)	−0.9914 (1.317)	0.4506 (1.450)
Share Hispanic	10.05	14.65	9.482 (0.8420)	10.50 (0.8068)	10.01 (0.9040)	1.018 (1.166)	0.5236 (1.235)
Share college+	20.04	8.414	18.71 (0.4751)	19.72 (0.3978)	21.90 (0.6090)	1.015 (0.6197)	3.190*** (0.7724)
Median HH income	50,306	12,751	48,866 (725.2)	51,285 (674.6)	50,444 (833.4)	2,418** (990.5)	1,577 (1,105)
Share in poverty	15.37	5.934	15.91 (0.3740)	14.76 (0.2862)	15.67 (0.3927)	−1.153** (0.4709)	−0.2422 (0.5423)
Density (housing units)	48.74	129.3	31.05 (4.968)	36.72 (3.536)	84.44 (12.95)	5.666 (6.098)	53.39*** (13.87)
Transit score	5.085	12.44	2.799 (0.5513)	4.128 (0.5381)	9.057 (1.093)	1.329* (0.7703)	6.258*** (1.224)
ICU beds per 100k	10.72	16.34	8.024 (0.8492)	10.37 (0.8283)	14.08 (1.171)	2.342** (1.186)	6.059*** (1.447)
Ever adopted SHO	0.7921	0.4060	0.7159 (0.02744)	0.7025 (0.02404)	1.000	−0.01339 (0.03648)	0.2841*** (0.02744)
N	890		271	363	256		

Notes. The table reports triplet counties’ summary statistics for mobility measures (top six rows) as well as relevant demographic, economic, and geographic characteristics of sample counties (next eleven rows). Mobility measures are computed for March–April 2019 to estimate typical pre-SHO mobility during the time of year in our main sample. Numbers of visits are per 100 people of the destination county’s 2018 population. Shares are in percent. “Ever adopted SHO” is an indicator for whether the county adopted an SHO at any point during March–April 2020. It is equal to one for all treated counties by definition. The table reports the sample average and standard deviation for all counties in the triplets. It also reports conditional means for each county type, as well as differences in sample means. Robust standard errors are in parentheses. Median HH Income refers to “Median Household Income.”

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

implementation of the SHO in the treated county. For example, if all treated counties in a given state implemented the SHO at the same time because of state-level mandates, all triplets formed around these treated counties would be placed in the same cluster. If counties at the southern border of the state implemented at a different date than counties along the northern border of the state, the two sets of resulting triplets would be placed in two different clusters. This baseline clustering criterion is motivated by the fact that many treated counties lie along state borders. The partition of the final sample of triplets results in 51 clusters. We called this “clustering by state and SHO implementation date” or by “state

and treatment date.” Results with alternative clustering schemes and alternative methods of estimating clustered standard errors are presented in Online Appendix A.2.7.

We also conduct an event study to assess pretrends and the dynamics of the SHO adoption impact. We use the following difference-in-difference dynamic specification:

$$y_{c(pr)t} = \psi_{c(pr)} + \sum_{s \in S - \{-1\}} \eta_s D_{p(t+s)} + \sum_{s \in S - \{-1\}} \beta_{T,s} (T_{pr} \times D_{p(t+s)}) + \sum_{s \in S - \{-1\}} \beta_{N,s} (N_{pr} \times D_{p(t+s)}) + X_{c(pr)t} + \epsilon_{prt}, \quad (3)$$

where $s \in [-4, 3]$ indexes the day relative to the treated

county's treatment date, with the noninteracted $D_{p(t-1)}$ dummies at $t - 1$ omitted. The definition of all other variables as well as clustering and weighting is consistent with the main specification in (2).

4.2. Identifying the Mechanisms

When a treated county implements an SHO, it can affect the mobility and social interactions in the neighbor county through two channels. First, the SHO can affect visits to establishments in the neighbor county originating in the treated county, as treated county residents alter their mobility in response to the SHO. The sign of this effect, which we call the *external traffic channel*, is a priori ambiguous. On the one hand, the closure of establishments in the treated county may lead its residents to seek alternatives nearby, increasing visits to the neighbor county. On the other hand, if residents of the treated county decide to follow the order, they would visit fewer establishments within their own county and the neighboring ones. Second, the SHO can also induce changes in internal mobility in the neighbor county. Residents of the neighboring county, in fact, may reduce their mobility by voluntary social distancing as a reaction to learning about the SHO implemented in the treated county. We label this effect the *internal traffic channel*, and assume that it may be driven by information transmission and sharing, consistent with models of epidemic transmission in which voluntary social distancing is driven by information diffusion (Eichenbaum et al. 2021). Ex ante, we expect the sign of this second effect to be in line with the direct effect experienced in the treated county—for example, because SHOs raise the neighbor county residents' awareness about the severity of the pandemic.

We quantify the relative importance of these channels in driving the estimated spillovers and investigate our conjecture about the importance of information sharing for the internal traffic channel by estimating two additional regression specifications. First, to test the hypothesis that any decline in internal mobility may be driven by information factors, we split the triplet sample in two and reestimate the specification in Equation (2). The first subsample consists of triplets in which the treated and the neighbor counties share the same local television news networks because they are in the same DMA.¹⁶ Counties in the same media market share exposure to common media sources that can shape how they react to neighboring SHOs (Burszty et al. 2020). We then repeat the above analysis on each subsample and compare the results. If a decline in internal traffic is coming about because of information transmission and sharing, we should observe stronger spillovers in the sample where treated and neighbor counties are in the same DMA.

Second, to establish the signs of spillovers through external and internal traffic channels and quantify their relative importance, we analyze directional traffic flows. The DMA analysis enables us to establish the presence

of an information transmission mechanism that may materialize through the reduction of internal mobility. In this second set of analyses, our aim is to quantify how much of the total mobility spillover to the neighbor county is driven by the internal traffic channel and how much of the reduction is due to the external traffic channel. To conduct this analysis, we use the mobility data between counties discussed in Section 3. We focus on traffic flows between counties in the triplet and use the same contiguous county identification strategy as in the estimation of spillover effects, replacing the outcome variable with measures of external and internal traffic. As in the main specification, we weigh observations by total population in the triplet as recorded in the 2018 American Community Survey. We cluster the standard errors by state and SHO implementation date as in the estimation of Equation (2).

We proceed in two steps. First, we calculate the total daily internal traffic for each county in the triplet. A county's internal traffic is defined as the total daily visits to establishments located inside the county's borders coming from devices that also reside in that county. Using this outcome variable, the interaction between the *treated* county identifier T_{pr} and the posttreatment indicator $Post_{pt}$ measures the effect of the treated county SHO on visits coming exclusively from its own residents. The interaction between the *neighbor* county identifier N_{pr} and the posttreatment indicator $Post_{pt}$ measures the change in internal traffic in the neighbor county as a result of the SHO of the treated county. Because establishments and residents of the neighbor county are not mandated to reduce mobility by the treated county's SHO, any change in internal visits in the neighbor county can be interpreted as voluntary, possibly driven by learning about the nearby SHO, for example through exposure to news and media coverage of the SHO in the treated county. We call this the internal traffic channel of spillovers.

Second, we calculate total visits to the neighbor county coming from the treated and hinterland counties separately in each triplet. For each triplet and day, there are now two observations, corresponding to visits of locations in the neighbor county by residents of treated and hinterland counties, respectively. With these variables, we estimate a difference-in-differences specification with one treated unit (traffic from treated to neighbor county) and one control unit (traffic from hinterland to neighbor county):

$$y_{c(pr)t} = \psi_{c(pr)} + \eta Post_{pt} + \beta_T (T_{pr} \times Post_{pt}) + X_{c(pr)t} + \epsilon_{prt}, \quad (4)$$

The interaction between the treatment county identifier T_{pr} and the posttreatment indicator $Post_{pt}$ measures the change in traffic coming from the treated county to the neighbor county, relative to the change in traffic from

the hinterland to the neighbor county. This is our measure of the spillover manifested through changes in external traffic, and we call it the external traffic channel.

One challenge encountered in the directional analysis stems from observations with zero traffic reported when considering counties with low traffic between them in any given day. The observed instances of zero traffic are mostly coming from traffic flows from the hinterland to neighbor counties. No *measured* external traffic does not necessarily imply no visits taking place. First, recall we use a sample of the total universe of devices, so there may be out-of-sample devices traveling between these counties. Second, to preserve privacy, SafeGraph removes recorded daily visits between a pair of CBGs when they are below four. This could result in zero traffic between counties when aggregating the corresponding CBGs. To address this issue, we first drop the triplets where we observe zero visits from our full sample when considering this directional traffic flow analysis, leaving us with 427 triplets. Although this only affects our second outcome variable, the external traffic, we analyze internal and external traffic with the same set of triplets for consistency. As a robustness check, we recode triplets with fewer than five visits as having four visits instead of dropping them.

5. Results

In this section, we report all estimation results. First, we discuss the estimated direct and spillover effects of staggered SHO implementation based on the contiguous counties regression design specified in Equation (2). Next, we explore the transmission mechanisms.

5.1. Estimated Direct and Spillover Effects

Table 2 reports the results from the estimation of our main specification that evaluates the mean change in the outcome variables between the preperiod (−4 to −1 days before the treated county’s SHO) and the

postperiod (0 to 3 days after). The coefficients of interest are β_T and β_N in Equation (2). They measure the direct and spillover effects on the outcomes of interest, on the treated and the neighbor counties, respectively, caused by SHOs implemented in the treated county in the triplet p , relative to the hinterland county that serves as control.

In the treated county, an SHO leads to lower mobility and interactions as measured by total visits to nonmedical establishments, more time spent at home, a higher fraction of devices all day at home, and a smaller fraction of devices at work. Visits to restaurants and retail establishments also decline. These effects are precisely estimated and have the expected sign. On average, treated counties experience an 11.8% decline in total visits during the four days after the SHO implementation.¹⁷

The coefficient η on the *Post* dummy in Equation (2) captures the change in hinterland county mobility in the four days following the implementation of the policy. In Table 2, this coefficient is statistically insignificant for all outcomes except for hours at home, which increase almost a full hour after the SHO takes effect relative to the period before the intervention. The positive coefficient on the *Post* dummy for hours at home indicates that residents in hinterland counties spent almost an hour more at home after the SHO, but residents of treated counties spent an additional half hour more at home relative to residents of hinterland counties. Overall, these findings provide strong evidence that the post-treatment difference between treated and hinterland counties is due to a strong response in the treated rather than in the hinterland counties.

The SHO spillover effects are sizable and statistically significant for all outcome variables, ranging from approximately a third to a half of the size of the corresponding direct effects.¹⁸ Importantly, the direction or sign of the spillover effects is the same as that of the direct effects for all outcomes. There is no evidence of “perverse” spillovers in the sense of an increase in

Table 2. SHO Direct and Spillover Effects: Contiguous County Estimation Results

	(1) Tot. visits	(2) Rest. visits	(3) Ret. visits	(4) Hrs home	(5) Frac. work	(6) Frac. comp. home
$Post_{pt}$	0.029 (0.043)	0.046 (0.055)	−0.007 (0.041)	0.892*** (0.202)	0.008 (0.005)	−0.002 (0.007)
$T_{pr} \times Post_{pt}$	−0.118*** (0.017)	−0.109*** (0.018)	−0.122*** (0.022)	0.548*** (0.115)	−0.013*** (0.002)	0.025*** (0.003)
$N_{pr} \times Post_{pt}$	−0.044*** (0.011)	−0.048*** (0.014)	−0.045*** (0.016)	0.297*** (0.078)	−0.005*** (0.001)	0.007*** (0.002)
No. of obs.	13,200	13,136	13,184	13,200	13,200	13,200
R^2	0.990	0.986	0.991	0.833	0.258	0.704
R^2 (within)	0.098	0.093	0.084	0.215	0.058	0.185

Notes. The table reports the results from the estimation of Equation (2) as described in the text. Coefficients on weather controls are not reported. Standard errors are clustered at the state and treatment date as discussed in Section 4. The window width is [−4,3] days around the treated county’s SHO implementation during March and April 2020. Restaurant and retail visit data are missing for a small number of county-days, explaining the lower number of observations in those outcomes. Tot. visits, total visits; Rest. Visits, visits to restaurants; Ret. visits, visits to retail locations; Hrs home, hours spent at home; Frac. work, fraction at work; Frac. Comp. home, fraction completely home.

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

mobility in neighbor counties in response to an SHO in the treated counties. The drop in mobility in neighbor counties implies that the net effect of the two channels spelled out above, the external and the internal channels, is a reduction in traffic. This result is important because, as we discuss in the last section of the paper and in the corresponding Online Appendix, it opens up the possibility that staggered implementation of SHOs might yield an overall mobility reduction that is larger than the one that would be achieved by simultaneous adoption on the *average* observed (under staggered adoption) implementation date.

5.2. Dynamic Effects of Staggered SHO Adoption
Turning to the dynamic effects of staggered adoption, Figure 4 plots the event study results for each outcome variable, based on the specification in Equation (3) and the sample of county triplets. Table A.3 in the Online Appendix reports the corresponding regression coefficients. The figure reports the estimated direct and spillover effects four days before and three days after the SHO adoption. The day before the announcement of SHOs is the omitted category. The estimated impacts are the coefficients from the regression of our outcome variables on the interaction between the county status in the triplet and seven different implementation dummy variables. Implementation dummies interacted with the treated county measure the direct effects (black, darker vertical bars in the figure). Implementation dummies

interacted with the neighbor county measure the spillover effects (gray, lighter vertical bars in the figure).

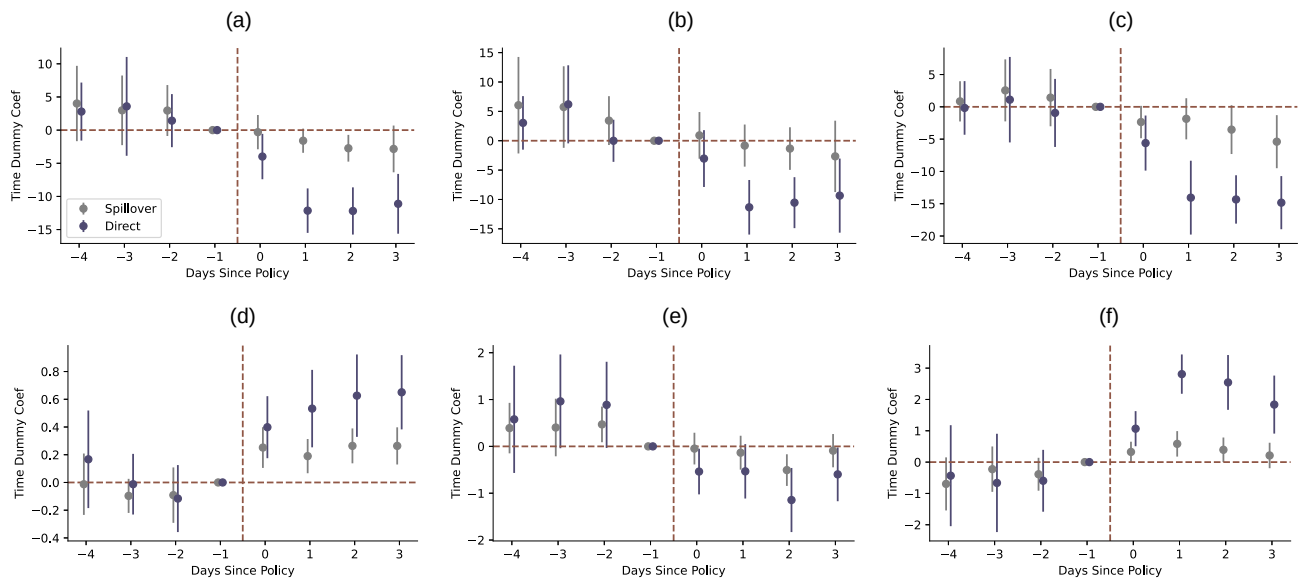
During the preperiod, the coefficients in the treated and neighbor counties are very similar, suggesting common mobility patterns relative to the hinterland (control) county. Although we do not see statistically significant differences in trends pre-SHO adoption across treated, neighbor and hinterland counties, some outcomes like restaurant visits and total visits seem to be declining somewhat in the preperiod in the treated and neighbor counties, consistent with the evidence in Berry et al. (2021). However, the figure also shows marked and statistically significant divergence from these pretrends after adoption.

Figure 4 also illustrates that after the SHO implementation, spillover effects are detectable in all outcomes and are statistically different from zero in all cases for at least one or two days after the implementation, except for restaurant visits. Both the direct and spillover effects are heterogeneous over time, building up in the first two days after adoption and, for some outcomes, leveling off or decreasing thereafter. Importantly, however, the estimated spillovers always run in the same direction as the direct effects for all outcomes.

5.3. Spillovers, Information Sharing, and Directional Mobility

It is clear from the estimation results that the spillover effects of county-based SHOs went in the same direction as the direct effects. What are the possible mechanisms

Figure 4. (Color online) SHO Direct and Spillover Effects: Event Study



Notes. (a) Total visits. (b) Restaurant visits. (c) Retail visits. (d) Hours at home. (e) Fraction at work. (f) Fraction completely at home. The figure plots estimated direct and spillover effects in event study format for all outcome variables. The x -axis is in days. Periods to the left of the dashed (red) line precede implementation, whereas periods to the right follow it; that is, $t = 0$ is the implementation day. The y -axis is in percent. Darker (navy) vertical bars are direct effects. Lighter (gray) vertical bars are spillover effects. The regression specification is given by (3)—it includes county fixed effects, as well as weather controls. Observations are weighted by county triplet population. $t = -1$ is the reference period. Coef, coefficient.

behind these changes in mobility? We now turn to unpacking the mechanisms behind the spillovers.

5.3.1. Spillovers and Local Media Markets One of the possible channels that could affect mobility in neighboring counties in response to a focal county SHO is an information signal, indicating that disease spread is significant enough to precipitate mobility restrictions. We conjecture that sharing a media market with the treated county, and hence receiving the same local news, can make residents of a neighbor county more aware of the treated county's SHO status, as well as the health risks leading to its adoption, inducing them to adopt *voluntarily* social distancing behaviors. DMAs are local media markets used to determine advertising targets and for other business purposes. The areas are determined by Nielsen, a major national marketing corporation, aggregating locations that share the same local TV stations and in which local broadcasting has a significant share of the total market in terms of total hours viewed. There are 210 DMA regions, covering the entire continental United States, Hawaii, and parts of Alaska. Despite the surge in online access to news, recent Pew Research Center surveys show that nearly 50% of households still get their news from local TV stations (Pew Research Center

2019). By considering county triplets whose treated and neighbor counties are located in the same DMA, we can control for such information-sharing channels of transmission. We split our triplet sample into those in which neighbor and treated counties share the same media market and those in which they do not.¹⁹ We end up with two subsamples of 310 and 240 triplets, respectively. We then run the same regression specified in Equation (2) for the two subsamples for all the outcome variables.

Table 3 reports the results for the two subsamples, same DMA and different DMA, for each outcome variable. For comparison, we also report the results for the full sample with all triplets. The direct effects are similar in the three samples for the total number of hours at home and the fraction of devices at work and at home (lower part of the table), whereas they are stronger for total visits, visits to restaurants, and to retail POIs in the common-DMA subsample. The direct effects always have the same sign and similar magnitude, in the different-DMA subsample.

In contrast, the spillovers are clearly driven by counties that share the same DMAs, consistent with the presence of an information channel for all outcomes except the fraction of devices at work. The spillover effects in

Table 3. Direct and Spillover Effects by Designated Market Area

	(1) All	(2) Same DMA	(3) Different DMAs	(4) All	(5) Same DMA	(6) Different DMAs	(7) All	(8) Same DMA	(9) Different DMAs
Log total visits				Log restaurant visits			Log retail visits		
$Post_{pt}$	0.029 (0.043)	0.101*** (0.027)	−0.116* (0.071)	0.046 (0.055)	0.137*** (0.031)	−0.140 (0.094)	−0.007 (0.041)	0.057** (0.026)	−0.136** (0.068)
$T_{pr} \times Post_{pt}$	−0.118*** (0.017)	−0.136*** (0.018)	−0.083*** (0.024)	−0.109*** (0.018)	−0.128*** (0.020)	−0.071*** (0.020)	−0.122*** (0.022)	−0.140*** (0.025)	−0.088*** (0.026)
$N_{pr} \times Post_{pt}$	−0.044*** (0.011)	−0.054*** (0.011)	−0.021 (0.024)	−0.048*** (0.014)	−0.056*** (0.016)	−0.029 (0.024)	−0.045*** (0.016)	−0.059*** (0.020)	−0.017 (0.022)
No. of obs.	13,200	7,440	5,760	13,136	7,416	5,720	13,184	7,440	5,744
No. of triplets	550	310	240	550	310	240	550	310	240
R^2	0.990	0.992	0.988	0.986	0.989	0.984	0.991	0.993	0.989
R^2 (within)	0.098	0.177	0.123	0.093	0.192	0.106	0.084	0.117	0.137
Hrs home				Frac work			Frac completely home		
$Post_{pt}$	0.892*** (0.202)	1.228*** (0.125)	0.211 (0.344)	0.008 (0.005)	0.014** (0.006)	−0.004 (0.007)	−0.002 (0.007)	−0.009 (0.006)	0.014 (0.009)
$T_{pr} \times Post_{pt}$	0.548*** (0.115)	0.650*** (0.128)	0.371*** (0.141)	−0.013*** (0.002)	−0.013*** (0.002)	−0.014*** (0.002)	0.025*** (0.003)	0.028*** (0.004)	0.020*** (0.002)
$N_{pr} \times Post_{pt}$	0.297*** (0.078)	0.402*** (0.100)	0.094* (0.057)	−0.005*** (0.001)	−0.005*** (0.001)	−0.006*** (0.002)	0.007*** (0.002)	0.009*** (0.002)	0.004 (0.003)
No. of obs.	13,200	7,440	5,760	13,200	7,440	5,760	13,200	7,440	5,760
No. of triplets	550	310	240	550	310	240	550	310	240
R^2	0.833	0.812	0.873	0.258	0.293	0.253	0.704	0.721	0.691
R^2 (within)	0.215	0.346	0.040	0.058	0.101	0.054	0.185	0.237	0.142

Notes. The table reports the results from the estimation of Equation (2), splitting the county triplet sample in two: one subsample that includes all triplets in which the treated and neighbor counties share the same DMAs, and one in which they don't. All regressions include county fixed effects as described in the text. Coefficients on weather controls are omitted from the table to conserve space. Standard errors clustered at the triplet level are in parentheses. Restaurant and retail visit data are missing for a small number of county-days, explaining the lower number of observations in those columns. The top half of column (1) in the table is the same as column (1) in Table 2.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

the different-DMA subsample are statistically insignificant for most outcome variables, with point estimates at least half of those in the same-DMA subsample. The total number of visits drops by a highly significant 5.4% when the neighbor is in the same DMA as the treated county. In sharp contrast, total visits drop by only a statistically insignificant 2.1% in the different-DMA subsample. Similar differences appear for restaurant visits and hours at home. The spillovers remain significant for the fraction of devices at home (marginally) and at work in the different-DMA subsample, but the point estimates are quantitatively comparable only for the fraction of devices at work, which might be less responsive to voluntary social distancing.

5.3.2. Spillovers and Directional Mobility. The differences in the strength of the spillover effects between DMA subsamples suggest that the spillovers may be due to residents in the neighbor county voluntarily restricting their mobility in response to the treated county's SHO. To further investigate this hypothesis, we quantify the relative contribution of the internal and external traffic to the total mobility spillover effects, using daily directional mobility data that take into account not just the location of the POIs but also the device's home location.

Table 4 reports the results of the estimation of the same specification in Equation (2) for three outcome variables: total traffic regardless of the origin that corresponds to the specification in our baseline results (column 1), total traffic originating within the same county (column 2), and total traffic from the treated and hinterland counties to the neighbor county, that is, external traffic to the neighbor county (column 3).

The first column of Table 4 reports results comparable to those in the first column of Table 2. The only difference relative to the baseline results is the slightly smaller triplet sample size. There are fewer observations than in the baseline sample because we omit county pairs that have days with zero traffic between them in the preperiod. We provide more detail and a robustness check at the end of this subsection. The estimated coefficients using this restricted sample are slightly different but consistent with the baseline results. As in the baseline, we find that the SHO in the adopting county reduces its total mobility. The reduction of traffic in the neighboring county is similarly strong, and the change in hinterland traffic, measured by the Post dummy, is similarly insignificant.

In the second column, the outcome variable is the total daily internal traffic for each county in the triplet. The estimated coefficients show strong evidence of negative direct and spillover effects on internal traffic following the treated county's SHO. The adoption of an SHO in the treated county reduces the county's own internal traffic by 4%. The spillover effect on the internal traffic of the

Table 4. SHO Impact on Internal and External Mobility of Neighbor Counties

	(1) All traffic	(2) Internal traffic	(3) External traffic
$Post_{pt}$	0.0278 (0.0586)	0.0105 (0.0164)	0.0225 (0.0499)
$N_{pr} \times Post_{pt}$	−0.0437** (0.0163)	−0.0197** (0.00877)	
$T_{pr} \times Post_{pt}$	−0.116*** (0.0235)	−0.0415*** (0.00936)	−0.0793*** (0.0191)
N	10,248	10,248	6,832
N triplets	427	427	427
R^2	0.990	0.998	0.827
R^2 (within)	0.0931	0.0523	0.00823
Weights	Yes	Yes	Yes

Notes. These regressions decompose the decrease in total visits to a county's POIs by origin of the visit. The analysis is restricted to county triplets in which there is at least one cross-county visit on each day of the sample. Both columns (1) and (2) follow the main specification in Equation (2). The dependent variable in column (1) is log total visits as in Table 2. The coefficients differ from those in column (1) of Table 2 because of the elimination of the triplets with zero external traffic. The dependent variable in column (2) is log total visits originating in the same county ("internal traffic"). The estimated coefficients indicate the decline in visits made by the residents in the treated county to its own establishments and from residents in the neighbor county to its own POIs, in response to the adoption of the SHO relative to the internal visit declines in the hinterland. Column (3) follows the specification in Equation (4), where the dependent variable is visits from the treated and hinterland counties in the triplet to the neighbor county. The sample excludes visits from the neighbor county because these are internal traffic to neighbor—each observation is a visit from the treated or hinterland county into the neighbor county—hence the blank spaces in the second set of table rows correspond to the interaction $N_{pr} \times Post_{pt}$. The estimated coefficient measures the decline in treated-to-neighbor visits after the SHO relative to the change in hinterland-to-neighbor visits. Standard errors in parentheses.

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

neighbor county is approximately half the size of the direct effect. Although they are not mandated by an SHO, residents of the neighbor county reduce their visits to the establishments in their own county after learning about the nearby SHO. This is direct evidence of the internal traffic channel at work: it's not just that residents of counties that implement SHOs stop visiting their neighbors, but the neighbor's own residents stay home more, voluntarily when adjacent counties implement SHOs. This evidence strongly corroborates the interpretation of our DMA results—an SHO in a county sends an information signal that affects behavior in neighboring counties.

In the third column, the outcome variable is the traffic from the treated or the hinterland counties to the neighbor county in their corresponding triplet, which we term the external traffic. This specification, presented in Equation (4), excludes the observations for the neighbor county in each triplet. Hence, the blank space in the second set of table rows corresponds to the interaction $N_{pr} \times Post_{pt}$. This is because, for the neighbor county,

“visits to the neighbor county” are internal visits that do not correspond to the external traffic flows that this specification aims to capture. The third set of rows in the table shows the estimate of the effect of the treated county’s SHO on the visits of its residents to the neighbor county, relative to the visits to the neighbor county originating in the hinterland county. The estimated coefficient measures the decline in treated-to-neighbor visits after the SHO relative to the change in hinterland-to-neighbor visits. The reported estimate implies a reduction of about 8% in external traffic from the adopting location. This is the impact of the external traffic channel on the overall spillover effect. Even if establishments remain open in the neighbor county, they experience a reduction in visits from residents of the county that adopted the SHO relative to visits from the hinterland. This result is intuitive because in Table 2, we document that SHOs induce not only fewer total visits, but also more time at home, less time at work, and the fraction of people completely at home that unambiguously implies less mobility from the implementing county to the neighbor.

The percentage decline of external visits from treated neighbor counties is around twice as large as the decline in the destination county’s own internal visits. But to compare these magnitudes, one must take into account the fact that most visits to a county’s POIs are internal. For the median county, 85% of the visits to its POIs come from its own residents. The total spillover effect is $(-0.0793)(1 - 0.85) + (-0.0197)(0.85) = -0.02864$. Of this 2.8% decline, the external traffic channel contributes $(0.0793)(1 - 0.85) = 1.189\%$, whereas the internal traffic contributes $(0.0197)(0.85) = 1.674\%$ —which is 41% more. Both channels of transmission are present in the data, but the internal traffic component of the total spillover effect is much more important in relative terms, consistent with the evidence of information sharing and transmission from the DMA analysis reported earlier. We conclude from these two sets of estimates that voluntary social distancing through information transmission is a quantitatively important driver of the spillover effects that we estimate, consistent with behavioral epidemiological models and models of voluntary social distancing.

As previously discussed, one challenge here is that, on some days, we observe zero external traffic to the neighbor county, mostly in the traffic flows from the hinterland to neighbor counties. However, no *measured* external traffic does not necessarily imply that there are no visits between these counties in the triplet.

Additionally, when daily directional visits are below four, SafeGraph removes these individual devices from the CBG count. To address this issue, we first drop the triplets in which we observe zero visits from our full sample, leaving us with 427 triplets. Although this only affects our second outcome variable, the external traffic,

we perform the analysis for both internal and external traffic with the same set of triplets for consistency. For comparability, we also reproduce and report the baseline results from Section 4.1 using this reduced set of triplets. Second, as an additional robustness check, we reestimate the same model retaining the full sample of counties, but recoding triplets with fewer than five visits as having four visits (as opposed to dropping them). The results, reported in Table A.4 in the Online Appendix, are very similar.

5.4. Robustness

The critical identification assumptions of our empirical analysis are embedded in the selection criteria for the county triplets in the final sample. In this section, we consider alternative criteria to construct our final sample to ensure that the results are not significantly affected. Our inference also relies on the precision of the estimated standard errors. A second important dimension along which we check the robustness of our analysis considers alternative clustering criteria. This section summarizes the results, whereas Online Appendix A.2 discusses the details and reports the results.

First, we assess the robustness of our results with respect to our choice of the time period around the SHO implementation date in the triplets. In the baseline specification, we impose the condition that neighbor counties in a candidate triplet do not adopt their own SHO for at least four days after the treated county does. If we imposed a three-day gap, we would have a sample with more triplets but a shorter panel of posttreatment days in which we can assess spillovers and higher risk of contamination from previous treatment in the neighbor county. If we imposed a five-day gap, we would have a sample with fewer triplets and a longer panel of posttreatment days with a lower risk of contamination from previous treatment. Online Appendix Tables A.6 and A.7 show that the sign and statistical significance of the estimated spillover effects remain unchanged in both settings.

Second, although our focus on contiguous county triplets nonparametrically controls for differential regional trends in mobility and county fixed effects control for permanent differences between counties *within* a triplet, our results could still be contaminated by time-varying differences between treated and hinterland counties, within the same triplet. Indeed, Table 1 suggests that neighbor and treated counties may be different from hinterland counties in some dimensions, which could possibly affect not only the level but also the trend in mobility. To address this concern, we group counties based on observable demographic and economic characteristics and restrict our sample to triplets where all counties are in the same homogeneous cluster. We group all U.S. counties into five clusters based on observables and keep only triplets where all counties are in

the same cluster, that is, all counties are “similar,” to the extent this is possible to gauge based on observables. Online Appendix Table A.8 reports the results of this robustness check. The approach leads to a considerably smaller sample of 121 triplets and weakens the power of our analysis, but the results are qualitatively robust and even become quantitatively stronger.

Third, to address the concern that same-DMA triplets are different from different-DMA triplets along some unobservable dimensions that may affect mobility, we rerun the media market analysis only on triplets in which the treated county belongs to at least one same-DMA and one different-DMA triplet. We call this sample “Overlapping Triplets.” In this specification, we test whether the spillover effects from the same treatment and the same county differ depending on whether the neighboring county is in the same DMA or not. The results in Table A.9 in the Online Appendix suggest that the different spillover effects in the two subsamples are not driven by unobservable dimensions along which the two subsamples of triplets differ. Despite the big loss in sample size, for most outcomes we document larger spillovers in neighbors that are part of the same DMA. This evidence strengthens our contention that when a county implements an SHO, the positive spillover effect on the mobility of surrounding locations is likely driven by information transmission and sharing. In Online Appendix A.2, we present additional evidence on the robustness of the results, including tests for heterogeneity of effects over time and within the subsample of triplets located in the same state. The results are not significantly affected across these robustness checks as well.

Finally, we run a comprehensive set of robustness checks on how we clustered the standard errors in the baseline analysis. We consider four alternative cluster criteria and two estimation methods to address potential inconsistencies in specifications with a small number of clusters. Table A.10 in the Online Appendix summarizes the results. The significance of the spillover effects on restaurant and retail visits is somewhat sensitive to the clustering methodology. However, the estimates of all other spillover effects are robust.

6. Health Policy Implications

In this section, we discuss the health policy implications of our empirical analysis and findings. We first illustrate that there are *two* relevant counterfactuals to the staggered adoption of SHOs that we witnessed in the United States during the first phase of the COVID epidemic. We then discuss their evaluation in terms of overall mobility reduction and argue that staggered adoption could potentially lead to a greater drop in mobility than a *delayed* coordinated intervention. This section also further clarifies the critical difference between the analysis

of contemporaneous and staggered spillovers, as estimated in this paper.

6.1. Alternative Policy Counterfactuals

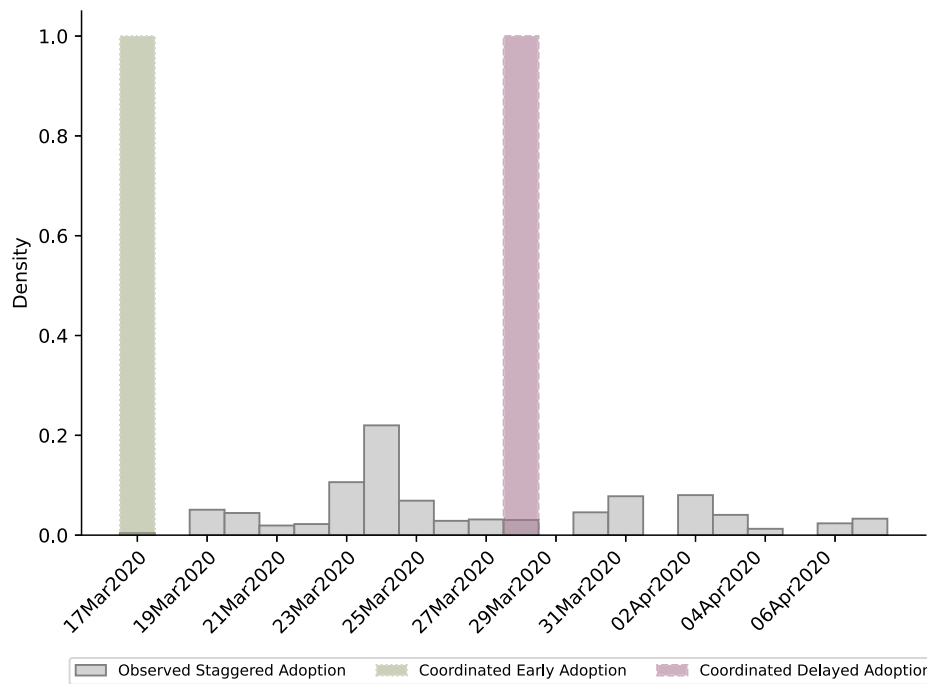
In the United States, the staggered implementation of COVID-19 SHOs has been criticized for potentially detracting from the mobility reduction that could have been obtained with a more coordinated policy intervention. A more coordinated SHO adoption at the *onset* of the pandemic would probably have promoted greater social distancing earlier on and likely led to fewer cases, hospitalizations, and deaths. However, that counterfactual confounds *coordinated* adoption with *earlier* adoption.

To see this, consider a mean-preserving reduction in adoption times: a situation in which SHOs are implemented with less time in between, but where the average implementation date is maintained as in the data. In this situation, counties that implemented SHOs earlier, like San Bernardino County in California, where measures were taken on March 19, would implement closer to the average implementation date of March 28. Counties that implemented later, like Ozark County in Missouri with an SHO implemented on April 6, would implement earlier.

Figure 5 illustrates the three alternative scenarios that one can consider. It plots the observed distribution of SHO implementation dates, which we call staggered adoption (SA; gray low bars), the counterfactual coordinated early adoption (CEA; green, lighter bar), and a coordinated delayed adoption (CDA; purple, darker bar). The CEA bar indicates the distribution that would have been realized had all counties that implemented an SHO during our sample period done so on March 17, when Henry County, GA, did. This counterfactual entails and may confound early adoption and coordination. The CDA bar shows the policy that could have been observed had all counties implemented their SHOs at the mean adoption date of March 28. This is a counterfactual that separates the mobility gains from early adoption from those possibly arising from coordination.

Our evidence of spillovers working in the same direction as the direct effects of SHOs (i.e., positive spillovers) opens up the possibility that the SA observed in the data may be preferable to a CDA counterfactual as in Figure 5 in terms of mobility reduction through the additional waves of indirect treatment brought about by the positive staggered spillovers. In the rest of this section, we show that this can indeed be the case, both in theory and in the data, based on a simple calibrated model and simulations based on the estimated version of Equation (2) as reported in Table 2.

Our study intentionally selects a smaller sample of triplets to enhance identification, accepting some limitations in terms of external validity. Thus, the findings should be viewed as applying to our particular triplet

Figure 5. (Color online) Staggered SHO Adoption and Alternative Counterfactuals

Notes. The figure plots the observed distribution of SHO adoption dates in our sample of U.S. counties (gray, lower bars, which we call Staggered Adoption or SA) and two counterfactual scenarios that we call Coordinated Early Adoption (CEA; the lighter, green bar) and Coordinated Delayed Adoption (CDA; the darker, purple bar). SA illustrates the actual distribution of SHO implementation dates. CEA shows a counterfactual in which all counties adopt on the earliest observed adoption date (March 17). CDA shows a counterfactual in which all counties adopt on the mean adoption date in the data (March 28). Unlike Figure 1, which plots the distribution for all U.S. counties that adopted an SHO during the early phase of the pandemic, this figure restricts the sample to treated and neighbor counties in our final sample of triplets that adopted an SHO between March 17 and on or before April 7. There are 782 such counties.

sample rather than generally applicable to all U.S. counties.

6.2. Comparing SA and CDA in a Simple Calibrated Model

In Online Appendix 10, we investigate under which conditions SA can potentially achieve a greater reduction in mobility than CDA by using a simple model that is calibrated using the estimated value of the staggered spillover effect from our estimates. In the model, there are three time periods, $t = 0, 1, 2$, and two counties, A and B. In each county, authorities can seek to reduce mobility by adopting an SHO at any time. However, their decisions will also affect mobility in the other county. Counties can either *simultaneously* adopt SHOs at $t = 1$ or they can *stagger* their adoptions, with A adopting at $t = 0$ and B adopting at $t = 2$ or, without loss of generality, vice versa.

If an SHO in county A increases mobility in county B (i.e., if the spillovers are “negative”), it is evident that simultaneous adoption always achieves greater mobility reduction regardless of whether it is delayed or not. But if spillovers are positive, as we find in the data, which policy is more effective depends on two factors, summarized by two parameters in our model.

The first, labeled β , captures the extent to which policymakers may value early mobility reductions more than later ones. Because mobility and contacts fuel the spread of the disease nonlinearly in standard epidemiological models, a given reduction in traffic today may be 5% more valuable than a reduction in traffic tomorrow. If that’s the case, we would assign $\beta = 1.05$.

The second, labeled λ , captures the strength of a spillover experienced by an already-treated county relative to an untreated one. For example, if county A adopts an SHO before county B, county B experiences a z reduction in traffic as a spillover effect. When county B adopts its own SHO, county A may also experience a spillover effect. But because county A’s SHO is already in place, this spillover could be weaker; that is, λz with $\lambda \leq 1$.

Our empirical strategy yields an estimate for z but not for λ and β .²⁰ To proceed, we consider a range of plausible values for both parameters. The analysis of the model in Online Appendix 10 shows that the mobility reduction advantage of SA over CDA decreases in both β and λ , and that the higher the parameter β or λ , the bigger the positive spillover $z > 0$ needs to be for SA to possibly dominate CDA. Next, by assigning a value to $\hat{z}$ based on the estimates in our data, we show that SA could dominate CDA for a wide range of λ and β values. For

example, even assuming that the policymaker places a 30% premium on next-period mobility, SA might still dominate CDA in terms of mobility reduction as long as spillovers experienced by counties with their own SHO already in place are less than two-thirds as strong as those without.²¹ Finally, we show that, if policymakers weigh time periods equally ($\beta = 1$), staggered adoption is preferable for *any* $\lambda \leq 1$.

6.3. An Empirical Counterfactual Analysis

The model above illustrates the policy implications of our empirical analysis by formally analyzing the relationship between the direction of spillover effects and the relative ranking of staggered versus coordinated adoption. It does so in a stylized two-county, three-period setting. In this section, we simulate counterfactual mobility under alternative coordinated adoption scenarios relying only on our empirical estimates and a minimal set of assumptions informed by the stylized model. The goal of the exercise is to quantify how much more (or less) aggregate mobility there could potentially have been during the early phase of the pandemic, assuming that counties in our triplets sample had coordinated on common implementation dates relative to the SA scenario we observed in the data, where “coordinated” here means simultaneous or contemporaneous county implementation. We consider the full range of possible coordinated adoption dates, with particular emphasis on the average implementation date, which represents the CDA scenario, and the earliest possible adoption date, which represents the CEA scenario, as in Figure 5. One caveat of our analysis is that it relies on estimates based on data from a subsample of county triplets with SHO implementation dates spread over only a few weeks.

As inputs to this analysis, we need (i) the set of estimated parameters from Equation (2) reported in Table 2, (ii) the actual SHO adoption dates, (iii) the shares of counties adopting on each date, (iv) whether they are treated or neighbor counties in the triplet, and (v) a minimal set of assumptions consistent with those made in the model. Online Appendix A.4 reports details on the construction of the three scenarios and the calculation of counterfactual mobility gains. We compare mobility across all counties in our final sample (as observed in the actual SA regime) to counterfactual mobility in the same set of counties under coordinated SHO adoption on *each* day from March 17 to April 7, including the *earliest* possible coordination day and the *average* implementation date, which correspond to the CEA and the CDA bars in Figure 5, respectively.

In each counterfactual, we consider what the total mobility across the entire sample period would have been had the SHO been adopted in all counties on the given day, and then plot the percent difference between

this counterfactual and the observed mobility in the SA as in Equations (A2) and (A3) in Online Appendix 10.

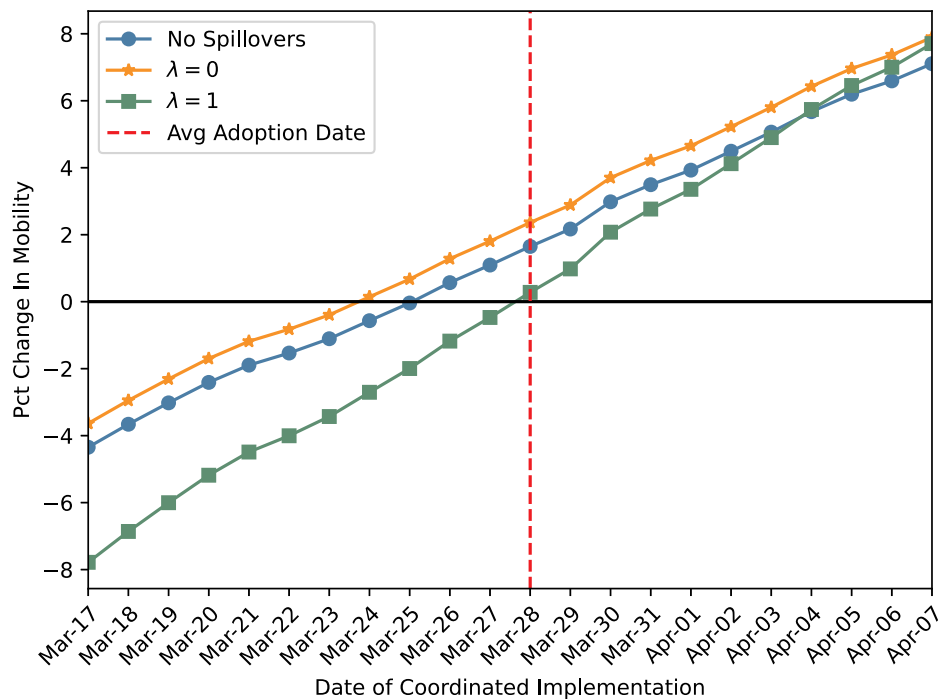
We first consider a case where counties only experience direct effects; that is, we assume that both changes in observed mobility and in counterfactual mobility under alternative coordinated adoption dates occur only because of SHOs in the adopting county. This case is labeled “No Spillovers” and is represented by the blue line with round markers. We then contrast this case with two other cases that assume spillovers of different intensities.

Figure 6 plots the results. Consider first the no-spillover case (blue line with round markers). As expected, the difference in mobility is increasing in adoption time regardless of the assumption on spillovers. In other words, the later the date of coordinated implementation, the greater the mobility in the coordinated scenarios and the better the outcome under SA. If all counties had shut down on March 17, when the first county in our sample did (the CEA scenario in Figure 5), aggregate mobility over the sample would have been between 3% and 8% lower than in the observed staggered adoption scenario. However, if all counties had waited until the last day in our sample—April 7—to shut down, mobility would have been 6%–8% higher. The reason why SA can still do better than DCA after a certain date even in the absence of any spillover is that, if DCA is adopted late enough, the cumulative mobility decline associated with it, by construction, is smaller than under SA.²²

Consider now the other two cases with spillovers. Our empirical analysis identifies and provides an estimate of β_N , the spillover effect experienced by counties before they implemented their own SHO. In the data, counties may also experience a spillover effect after they implement their own SHO, if a neighbor implements an SHO after the focal county. Moreover, in the coordinated adoption counterfactuals, all counties may experience spillover effects from their neighbors adopting at the same time. In the model, the effect of these “postspillovers” is controlled by $\lambda\beta_N$, where λ represents the relative strength of the unobserved postspillover relative to the observed spillover β_N , which for the purpose of this discussion we call “prespillover.” In Figure 6, we consider two polar cases in which either there are no “postspillovers” ($\lambda = 0$, orange line with star markers) or the “postspillovers” are as strong as prespillovers ($\lambda = 1$, green line with square markers).

The threshold date on which SA first delivers larger mobility reductions than coordinated adoption depends on the spillover effects. The difference between $\lambda = 0$ and “No Spillovers” lines isolates the effect of prespillovers. They unambiguously weaken the case for coordinated adoption. If counties coordinate on a common date, they effectively miss out on additional rounds of treatment that materialize when neighbors shut down first. The difference between the $\lambda = 0$ and

Figure 6. (Color online) Coordinated Adoption Time and Mobility Gains from SHO Implementation



Notes. The figure plots the percent difference in mobility reduction under coordinated adoption relative to that under SA, calculated as in Equations A2 and A3 in the Online Appendix (*y*-axes), varying the date of coordinated adoption (*x*-axes), under alternative assumptions on the spillover effects. A positive value on the lines means that mobility is *greater* in the coordinated counterfactual than the SA; that is, the observed SA achieves greater mobility reduction. The “No Spillovers” case sets $\beta_N = 0$. The other two cases use the spillover effect estimated in the data but illustrate the two polar cases of spillover intensity received by already-treated counties. As in our model, $\lambda = 0$ means no spillover of this kind. $\lambda = 1$ means maximum spillover of this kind. The mean adoption date, at which CDA occurs, March 28, is represented by the dashed vertical line.

$\lambda = 1$ lines isolates the effect of the postspillovers. When neighbors’ policy can further reduce mobility beyond the focal county’s own SHO, coordinated adoption is more effective than when this channel is shut off.

Of particular interest are the percent changes in mobility on March 28, the average adoption date in our sample. It represents the CDA scenario in Figure 5, which separates the effect of coordination from the effect of early adoption and is formalized in the model above. We can see that regardless of the strength of postspillovers, a counterfactual in which all counties had shut down simultaneously on March 28 would have led to 0.3%–2.3% more mobility than what we actually observed in the SA.

In conclusion, this set of alternative counterfactual scenarios clarifies and quantifies the sources of the gains from SHO-induced mobility reductions. It is clear that an early country-wide coordinated implementation of SHOs would have led to less mobility in the United States. But a coordinated implementation on a “compromise” date equal to the average implementation date in the data might not have achieved the same result. This clearly illustrates the fact that coordination in itself is not sufficient to maximize mobility

reduction. Early adoption is also required to achieve that goal.

7. Conclusions

In this paper, we estimate the direct and spillover effects of staggered implementation of SHO policies on smartphone-based mobility indicators. To identify these, we implement a difference-in-difference research design based on contiguous county triplets in which SHO adoption in the treated county can spill over onto a neighboring county that has not adopted at least four days before and does not adopt for at least three days after the focal county implementation day, and mobility in both counties is assessed relative to a hinterland control county. We find that spillover effects from staggered SHO implementation have the same sign as the direct effects; that is, spillovers reduce mobility in neighboring counties as well as in the treated counties. These spillovers are substantial, between a half and a third of the estimated direct effects.

We also investigate the mechanisms behind these spillovers by estimating the same model in two subsamples of county triplets differentially exposed to SHO news and by quantifying the direct and spillover effects on directional mobility indicators. When we split the triplet

sample along the same or different designated digital market areas, we find that spillover effects are much stronger in the presence of information sharing as captured by differential exposure to common sources of local news in the same media market.

When we look at internal and external traffic separately, we find that a county's SHO keeps its residents at home. They do not go out in their own county, nor do they flock to neighboring counties, even if the destination has looser restrictions. The reduction in visits from the adopting county decreases the overall mobility of the neighboring counties. We call this the external traffic channel of spillovers. We also find that the residents of counties neighboring the SHO-treated one reduce their own internal traffic, even if they are not directly mandated to do so by an SHO. We call this the internal traffic channel of spillovers. When we decompose the total spillover effect, we find that the change in internal traffic contributes twice as much as the change in external traffic to the total reduction in mobility in the neighboring counties. This evidence corroborates the hypothesis that information sharing and transmission drive our baseline results and highlights the potential mobility reduction benefits of voluntary social distancing behaviors.

Taken together, this body of evidence is consistent with staggered SHO adoption promoting awareness of the pandemic's severity and signaling the desirability of social distancing in surrounding counties, even in the absence of binding mobility restrictions. This is in line with behavioral models of social distancing in which individuals respond to information about the risks of infection.

These empirical results have an important health policy implication. Based on both a simple calibrated model example and a set of counterfactual simulations based on our reduced-form estimates, we illustrate that in the presence of spillovers that go in the same direction as the direct effects of NPIs as we estimate in the data, staggered implementation can contribute to, rather than detract from, the overall goal of reducing mobility that is a proxy for contacts and interactions. Importantly, we also show that the impact of SHO policy depends not only on coordination but, critically, also on the timing of adoption because of the additional rounds of spillover treatment that can increase voluntary social distancing under staggered adoption.

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Endnotes

¹ We focus on SHOs because they were the most widely adopted nonpharmaceutical intervention policy with substantial variation in adoption both over time and across counties. Moreover, in states that "shut down" gradually, SHOs were the most restrictive and often the intervention of last resort. As a result, we would expect them to have the strongest direct effect but, relatively speaking, the weakest spillover effects, making our estimates of spillover effects a conservative lower bound.

² See <https://www.newsweek.com/majority-americans-believe-trump-should-enact-nationwide-stay-home-order-poll-says-1498179>.

³ For example, Zhao et al. (2021) and Holtz et al. (2020), which we discuss in detail in the next section.

⁴ See Gupta et al. (2020), Avery et al. (2020), Harapan et al. (2020), and Brodeur et al. (2021) for surveys of the broader COVID-19 literature.

⁵ More precisely, this paper estimates spillovers in mobility when one county has adopted an SHO and the other has not. It deliberately excludes cases in which both counties (or several neighboring counties) adopt at the same time and thus experience concurrent spillovers from each other's policies. In contrast, Holtz et al. (2020) also estimate mobility spillovers when counties adopt such a policy at the same time. The two studies rely on similar data and in part also on similar identification strategies. Both studies decompose the spillovers into a component arising from external traffic and another related to internal traffic driven by information sharing, in a social media network in Holtz et al. (2020) or due to exposure to common sources of news in the same media market in this paper, with findings largely consistent with each other. As in Holtz et al. (2020), in this paper, a model of policy adoption helps to draw the policy implications of the empirical analysis. In particular, Holtz et al. (2020) draw policy implications from the empirical analysis in a game-theoretic model of policy interaction.

⁶ The primary source of the data is publicly available at <https://www.safegraph.com>. At the time of writing, the academic version of the SafeGraph data set is distributed by <https://www.deweydata.io/>.

⁷ We describe our county sample selection process in Section 4.1.

⁸ Safegraph Core Places follows a multiprong approach to assign POIs to categories. It uses Google Places classification together with other open-source information in a deep neural network model, complementing this with human-expert labeling.

⁹ The origin of the visit is taken to be the device's home location. SafeGraph identifies the home location of a device with a 153-m × 153-m square in which the device spends nighttime hours (6 p.m.–7 a.m.) over a six-week period.

¹⁰ Keystone data are available at <https://github.com/Keystone-Strategy/COVID19-intervention-data>.

¹¹ Online Appendix 10 sets up a simple model of NPI adoption that nests both simultaneous and staggered adoption and hence helps to further illustrate the differences between the analysis and identification challenges of spillovers from contemporaneous and staggered implementation.

¹² When multiple candidate hinterland counties exist, we select the one that is least connected to the treated county using the connectedness measure discussed above (see more details below)

¹³ Transit scores come from the Center for Neighborhood Technology and measure how well a location is served by public transit on a scale from 0 to 10, where 10 indicates higher accessibility. This index incorporates information on scheduled bus, rail, or ferry services.

¹⁴ The socio-demographic characteristics are from the 2018 American Community Survey.

¹⁵ As Gardner (2022) shows, the stacked difference-in-differences estimator yields a weighted average of the average treatment effect on the treated unit if the parallel trends assumption is satisfied. However, in a balanced sample such as ours, which Wing et al. (2024) term “trimmed,” the weights are the same in each triplet; a population-weighted approach such as the one we adopt here is recommended for consistent estimation of the average treatment effect.

¹⁶ DMAs, often referred to as “media markets,” are geographic areas that share the same local TV network affiliates, newspapers, and other media. See more on DMAs below.

¹⁷ Recall that observations are weighted by the triplet population.

¹⁸ As summarized below, in Online Appendix A.2, we explore the robustness of these results by varying the event window and restricting the analysis to alternative subsets of triplets. The estimates in these robustness tests are in line with those reported in Table 2.

¹⁹ Note that the hinterland county is ignored by the criterion for this sample split. All but one of the hinterland counties we consider are in DMAs that are different from the treated and the neighbor county DMAs. Results from omitting the triplet in which the hinterland shares a DMA with the other two counties are essentially the same.

²⁰ A study that does imply an estimate for λ is Holtz et al. (2020), though a different sample and different empirical strategy do not permit a direct comparison.

²¹ This analysis nests staggered and simultaneous or contemporaneous spillovers. Furthermore, in Online Appendix 10 we consider the possibility that a neighbor’s simultaneous adoption moderates the direct effect of a county’s own SHO (parameterized by $\phi \leq 1$). In that case, SA can dominate CDA over an even wider region of the parameter space.

²² Consider the three-period example in our model without any spillovers, that is, $z = 0$. If DCA does not take place in period 0 or 1, any staggered implementation reduces mobility more than DCA in period 2.

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